

TRIGONOMETRY FORMULA SHEET

Complete Reference Guide & Values Table

1. Right-Angled Triangle Ratios (કાટકોણ ત્રિકોણના ગુણોત્તર)

$$\sin \theta = \frac{\text{સામેની બાજુ (Opposite)}}{\text{કર્ણ (Hypotenuse)}}$$

$$\operatorname{cosec} \theta = \frac{\text{કર્ણ (Hypotenuse)}}{\text{સામેની બાજુ (Opposite)}}$$

$$\cos \theta = \frac{\text{પાસેની બાજુ (Adjacent)}}{\text{કર્ણ (Hypotenuse)}}$$

$$\sec \theta = \frac{\text{કર્ણ (Hypotenuse)}}{\text{પાસેની બાજુ (Adjacent)}}$$

$$\tan \theta = \frac{\text{સામેની બાજુ (Opposite)}}{\text{પાસેની બાજુ (Adjacent)}}$$

$$\cot \theta = \frac{\text{પાસેની બાજુ (Adjacent)}}{\text{સામેની બાજુ (Opposite)}}$$

2. STANDARD ANGLE VALUES TABLE (0° – 90°)

Function / Angle	0° (0 rad)	30° (π/6)	45° (π/4)	60° (π/3)	90° (π/2)
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Undefined
$\operatorname{cosec} \theta$	Undefined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Undefined
$\cot \theta$	Undefined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

3. ESSENTIAL TRIGONOMETRIC IDENTITIES & FORMULAS

Reciprocal & Quotient Identities

- $\operatorname{cosec} \theta = 1 / \sin \theta$
- $\sec \theta = 1 / \cos \theta$
- $\cot \theta = 1 / \tan \theta$
- $\tan \theta = \sin \theta / \cos \theta$
- $\cot \theta = \cos \theta / \sin \theta$

Pythagorean Identities

- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

Even / Odd (Negative Angle)

- $\sin(-\theta) = -\sin \theta$, $\operatorname{cosec}(-\theta) = -\operatorname{cosec} \theta$
- $\cos(-\theta) = \cos \theta$, $\sec(-\theta) = \sec \theta$
- $\tan(-\theta) = -\tan \theta$, $\cot(-\theta) = -\cot \theta$

Cofunction Identities

- $\sin(90^\circ - \theta) = \cos \theta$
- $\cos(90^\circ - \theta) = \sin \theta$

Double-Angle & Half-Angle

- $\sin(2\theta) = 2 \sin \theta \cos \theta$
- $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2\cos^2 \theta - 1$

- $\tan(90^\circ - \theta) = \cot \theta$
- $\operatorname{cosec}(90^\circ - \theta) = \sec \theta$
- $\sec(90^\circ - \theta) = \operatorname{cosec} \theta$
- $\cot(90^\circ - \theta) = \tan \theta$

- $\tan(2\theta) = 2 \tan \theta / (1 - \tan^2 \theta)$
- $\sin(\theta/2) = \pm \sqrt{[(1 - \cos \theta)/2]}$
- $\cos(\theta/2) = \pm \sqrt{[(1 + \cos \theta)/2]}$

Sum & Difference Formulas

- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
- $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
- $\tan(A \pm B) = (\tan A \pm \tan B) / (1 \mp \tan A \tan B)$

Triangle Laws

- **Law of Sines:** $a / \sin A = b / \sin B = c / \sin C$
- **Law of Cosines:** $c^2 = a^2 + b^2 - 2ab \cos C$
- **Area:** $\text{Area} = \frac{1}{2} ab \sin C$

Product-to-Sum & Sum-to-Product Formulas

- $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$
- $\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$
- $\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$
- $\sin A + \sin B = 2 \sin((A+B)/2) \cos((A-B)/2)$
- $\sin A - \sin B = 2 \cos((A+B)/2) \sin((A-B)/2)$
- $\cos A + \cos B = 2 \cos((A+B)/2) \cos((A-B)/2)$
- $\cos A - \cos B = -2 \sin((A+B)/2) \sin((A-B)/2)$